

Code No: 118EQ**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD****B. Tech IV Year II Semester Examinations, June - 2018****SEMANTIC WEB AND SOCIAL NETWORKS****(Computer Science and Engineering)****Time: 3 hours****Max. Marks: 75****Note:** This question paper contains two parts A and B.

Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART - A**(25 Marks)**

- | | | |
|------|---|-----|
| 1.a) | What is artificial intelligence? | [2] |
| b) | What is solvable on the Web? | [3] |
| c) | Explain about HTML language. | [2] |
| d) | Give an overview of Web ontology language. | [3] |
| e) | What is e-learning? Explain. | [2] |
| f) | Explain about monotonic and nonmonotonic rules. | [3] |
| g) | What is degree centrality? | [2] |
| h) | Discuss about Swoogle. | [3] |
| i) | Define Jaccard-coefficient. | [2] |
| j) | Explain about Sesame. | [3] |

PART - B**(50 Marks)**

- | | | |
|------|--|-------|
| 2.a) | Explain how HTML limits the manipulation of information. | |
| b) | Discuss about the information age. | [5+5] |

OR

- | | | |
|------|---|-------|
| 3.a) | Discuss how the number of nodes on the Web creates computational complexity that limits the ability to develop logic proof systems. | |
| b) | Describe the semantic Web road map. | [5+5] |
| 4.a) | Explain basic elements of RDF language. | |
| b) | Give examples of ontology and OWL. | [5+5] |

OR

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|------|---|-------|
| 5.a) | Discuss about ontology methods, matching and mapping. | |
| b) | Explain sharing and merging. | [5+5] |
| 6.a) | Describe logic and inference. | |
| b) | Discuss about semantic Web applications and Web services. | [5+5] |

OR

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|------|---|-------|
| 7.a) | Explain about inference engines. | |
| b) | Discuss about enterprise application integration. | [5+5] |

8.a) Explain about latent semantic index search.

b) What is social network analysis? Explain.

[5+5]

OR

9.a) Discuss about semantic search technology.

b) What are personal networks? Explain.

[5+5]

10. Explain in detail about building Semantic Web applications with social network features.

[10]

OR

11. Explain in detail about Web-based networks.

[10]

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Code No: 118EQ**JAWAHARLAL NEHRU TECHNOLOGICAL UNIVERSITY HYDERABAD****B. Tech IV Year II Semester Examinations, April - 2018****SEMANTIC WEB AND SOCIAL NETWORKS****(Computer Science and Engineering)****Time: 3 hours****Max. Marks: 75****Note:** This question paper contains two parts A and B.

Part A is compulsory which carries 25 marks. Answer all questions in Part A. Part B consists of 5 Units. Answer any one full question from each unit. Each question carries 10 marks and may have a, b, c as sub questions.

PART - A**(25 Marks)**

- | | | |
|------|--------------------------------------|-----|
| 1.a) | What is machine intelligence? | [2] |
| b) | Explain in detail about ontology. | [3] |
| c) | Describe ontology libraries. | [2] |
| d) | Discuss about constructing ontology. | [3] |
| e) | What is descriptive logic? | [2] |
| f) | Explain about knowledge base. | [3] |
| g) | What is TAP? | [2] |
| h) | Explain about random graph model. | [3] |
| i) | What is snowballing technique? | [2] |
| j) | Explain about GraphUtil. | [3] |

PART - B**(50 Marks)**

- | | | |
|-----------|--|-------|
| 2.a) | Discuss about logic on the semantic Web. | |
| b) | Explain about reasoning with semantic networks. | [5+5] |
| OR | | |
| 3.a) | List the possible uses for small devices that may develop within the next 5 years. | |
| b) | Explain about the World Wide Web. | [5+5] |
| 4.a) | Explain about ontology "Spot" example. | |
| b) | Describe compatibility of OWL and RDF/RDFS. | [5+5] |
| OR | | |
| 5.a) | Discuss about ontology engineering and ontology development tools. | |
| b) | Briefly explain about ontology schema. | [5+5] |
| 6. | Explain about semantic search semantic bioinformatics. | [10] |
| OR | | |
| 7. | Explain in detail about logic, inference and rules. | [10] |

- 8.a) Discuss about Web search agents.
b) Explain about the global structure of networks.

[5+5]

OR

- 9.a) What is a search engine? Explain about search engines.
b) Describe development of social network analysis.

[5+5]

10. Explain the following:
a) The generic architecture of semantic Web applications
b) Web-based networks.

[5+5]

OR

11. Explain the following:
a) Elmo
b) Blogs and online communities.

[5+5]

---ooOoo---